Loan Default Prediction

Comprehensive report: EDA findings, modeling methodology, and predictive results

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58,645	14.24%	Gradient Boosting
Records analyzed	**Observed default rate**	**Selected model**
Clean sample; no missing values	Imbalanced target	Chosen by CV PR-AUC

Executive conclusion

The recommended model is Gradient Boosting.

- **It achieved 0.954 ROC-AUC and 0.872 PR-AUC on the independent test set.**
- **At the optimized threshold of 0.417, it reached 90.8% precision, 72.8% recall, and 0.808 F1.**

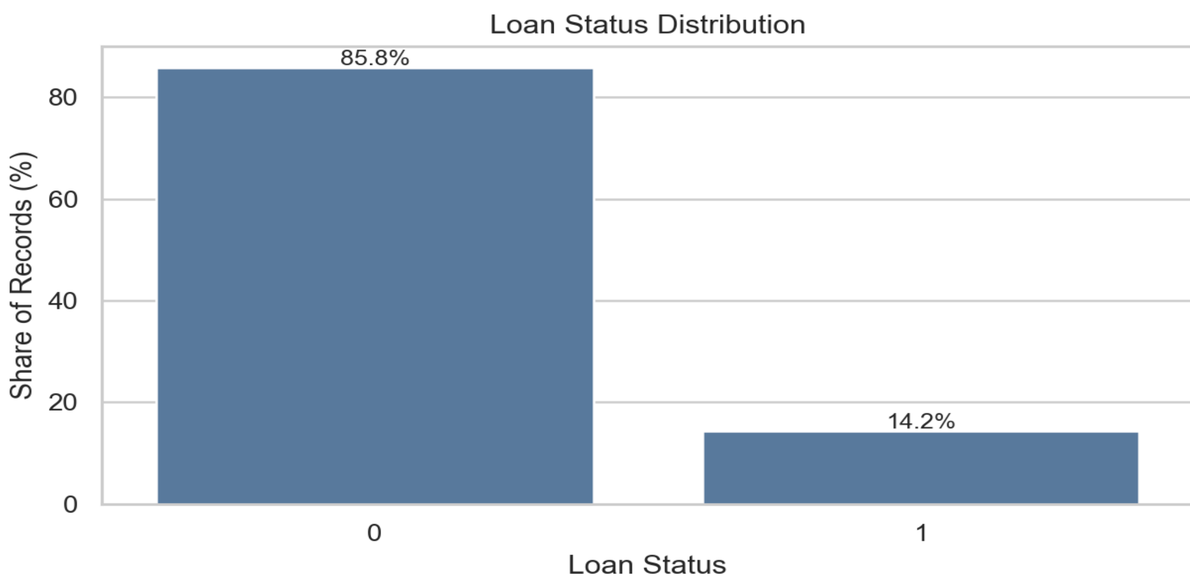
- Default risk is concentrated in repayment burden, pricing, housing status, income, loan intent, and loan grade.
- The target is imbalanced, so model selection is based on ranking metrics, especially PR-AUC, rather than accuracy alone.
- The final workflow avoids data leakage: cross-validation selects the model, out-of-fold predictions tune the threshold, and the test set is used only for final reporting.
- Operationally, the model should be used as a risk-ranking layer, with the final threshold selected based on review capacity and loss tolerance.

1. Dataset Structure and Data Quality

The dataset contains 58,645 observations and 11 eligible predictors (7 numeric, 4 categorical) after excluding the ID column, with no missing values or duplicate identifiers, indicating strong structural quality.

Check	Result	Readout
Rows / columns	58,645 / 13	Large enough for stable train-test evaluation
Independent variables	11 non-ID predictors	All eligible fields retained for modeling (7 Numeric + 4 Category)
Numerical predictors	7	Reviewed through distributions and correlation to default
Categorical predictors	4	Reviewed through category-level default rates
Missing values	0	No imputation needed
Duplicate IDs	0	No identifier duplication issue
Default observations	8,350(14.2%)	Minority class requires careful metrics

The target default rate is 14.2%, creating moderate class imbalance that requires evaluation metrics beyond simple accuracy.



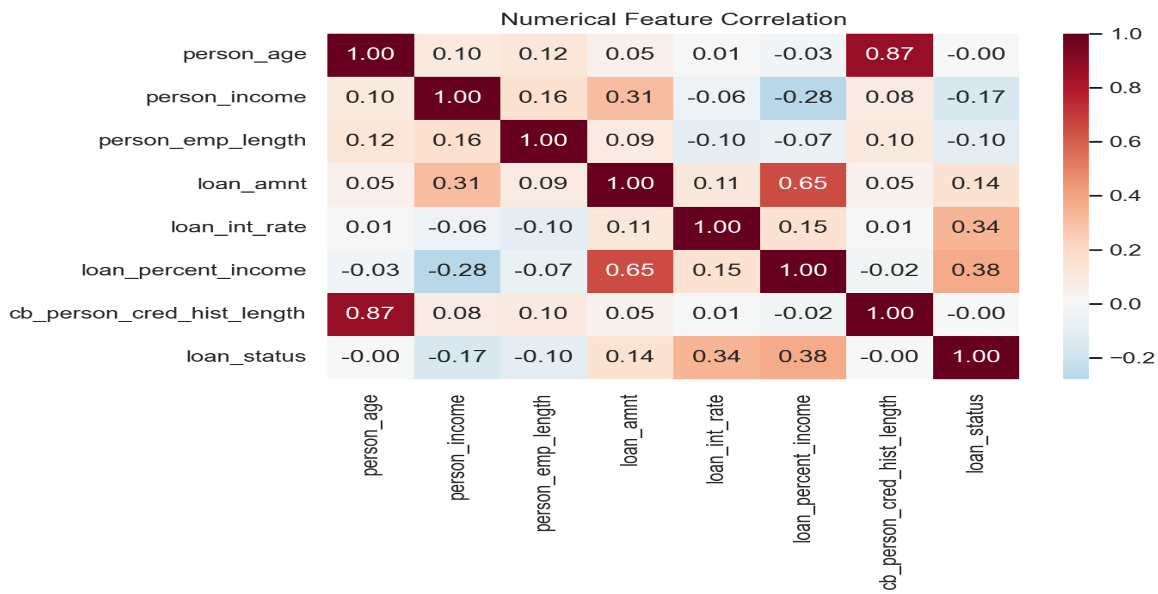
EDA implication: because defaults represent only 14.24% of the sample, accuracy can overstate model quality. Later evaluation therefore emphasizes ROC-AUC, PR-AUC, precision, recall, and F1.

2. EDA Findings - Numerical Risk Signals

Conclusion first:

- **Affordability (loan_percent_income) is the strongest numerical driver of default (corr ≈ 0.38),**
- **followed by interest rate (≈ 0.34), highlighting repayment burden and pricing as key risk signals.**
- **Higher income (≈ −0.17) and longer employment (≈ −0.10) are protective, while larger loan amounts show a weaker positive association (≈ 0.15).**

The heatmap provides the broad relationship view, while the table highlights the most decision-relevant numerical features for later modeling.



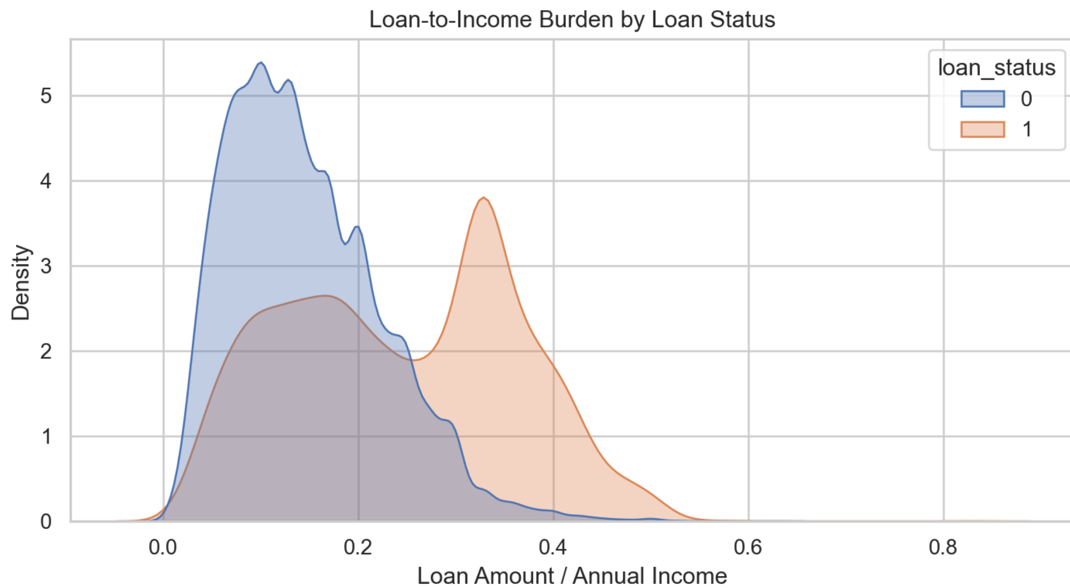
Feature	Correlation to default	Interpretation
loan_percent_income	0.378	Higher repayment burden
loan_int_rate	0.339	Risk-based pricing and repayment stress
person_income	-0.170	Higher income lowers observed risk
loan_amnt	0.145	Larger loans carry higher observed risk
person_emp_length	-0.100	Longer employment is modestly protective

* Notably, extreme maximum values (e.g., age or employment length equal to 123) may represent outliers and will be considered during modeling.

2. EDA Findings - Loan-to-Income Burden

Conclusion first:

- **Defaulted borrowers cluster around a loan-to-income ratio of ~0.30–0.40, versus ~0.10–0.20 for non-defaults.**
- **This aligns with its strongest correlation to default (≈0.38), well above loan amount (≈0.15).**



Senior readout: income matters, but affordability is more directly predictive. The practical risk question is whether the requested loan burden is sustainable relative to the borrower's income.

3. EDA Findings - Segment-Level Risk

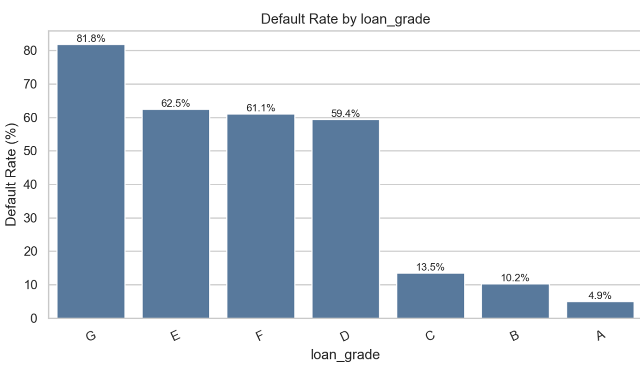
Conclusion first:

- **Loan grade shows the most pronounced separation in default rates and appears to contain ordinal risk information.**
- **person_home_ownership and cb_person_default_on_file also display meaningful differences, with prior default history being a strong logical risk indicator.**

Dimension	Highest-risk segment	Default rate	Lowest-risk segment	Default rate
Loan grade	G	81.82%	A	4.92%
Home ownership	RENT	22.26%	OWN	1.37%
Loan intent	DEBTCONSOLIDATION	18.93%	VENTURE	9.28%
Prior default file	Y	29.89%	N	11.51%

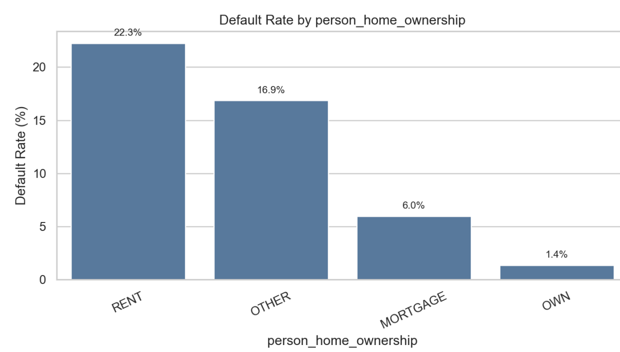
Loan grade:

lower grades show a steep default-risk gradient.



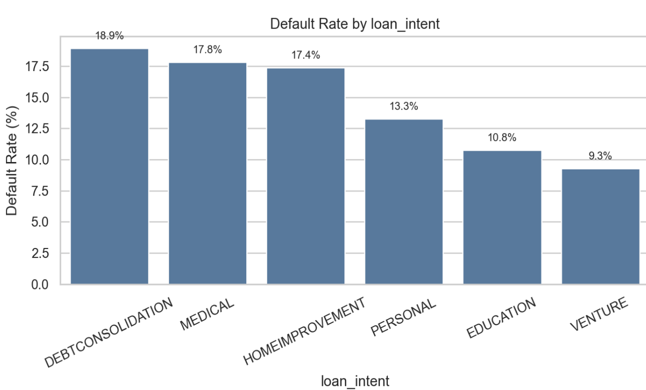
Home ownership:

renters carry materially higher observed risk.



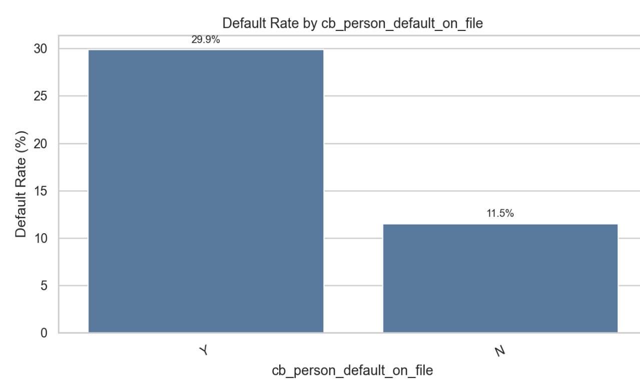
Loan intent:

debt consolidation and medical loans are higher risk.



Prior default:

prior default history materially lifts observed risk.



4. Modeling Methodology

The methodology strictly isolates the test set first and performs model selection (via CV PR-AUC) and threshold tuning (via OOF F1 optimization) entirely within training data, preventing leakage.

This ensures ranking quality is validated before fixing a decision threshold, aligning model evaluation with real-world credit decision logic.

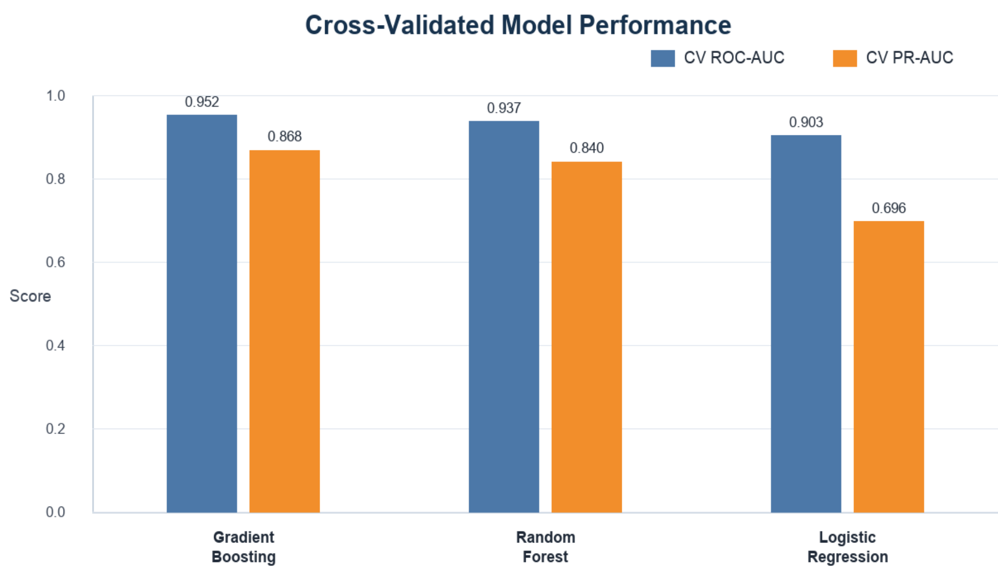
Step	Action	Boundary
1	Define target and predictors	loan_status is the target; id is excluded.
2	Create stratified train/test split	The holdout test set is isolated first.
3	Build preprocessing pipeline	Numerical and categorical variables are handled separately.
4	Compare candidate models	Logistic Regression, Random Forest, and Gradient Boosting are compared inside train_full.
5	Select model by CV PR-AUC	Gradient Boosting is selected based on cross-validated ranking quality.
6	Tune threshold with OOF predictions	The selected model's threshold is optimized to maximize F1.
7	Report final test performance	The untouched test set is used only once for final reporting.

Why this matters: CV PR-AUC evaluates ranking quality before any fixed classification threshold is chosen. Precision, recall, and F1 are then interpreted after threshold tuning, which better reflects an operational decision process.

5. Model Selection by Cross-Validation

Decision boundary: three candidate models were compared only inside train_full using 5-fold cross-validation. Based on CV PR-AUC, Gradient Boosting was selected. Only the selected model proceeds to threshold tuning and final test reporting.

Model	CV ROC-AUC	Std.	CV PR-AUC	Std.	Decision
Gradient Boosting	0.952	0.0038	0.868	0.0051	Selected
Random Forest	0.937	0.0021	0.840	0.0047	Benchmark
Logistic Regression	0.903	0.0035	0.696	0.0087	Benchmark



Readout: Gradient Boosting has the strongest PR-AUC and ROC-AUC, so the remaining threshold and test analysis is intentionally limited to that selected model.

6. Selected Model Threshold and Final Test Results

After model selection, the classification threshold is tuned with out-of-fold predictions from train_full. The tuned Gradient Boosting model is then evaluated once on the untouched test set.

Stage	Model	Threshold	Precision	Recall	F1	ROC-AUC	PR-AUC
OOF threshold tuning	Gradient Boosting	0.417	91.7%	73.2%	0.814	-	-
Final test result	Gradient Boosting	0.417	90.8%	72.8%	0.808	0.954	0.872

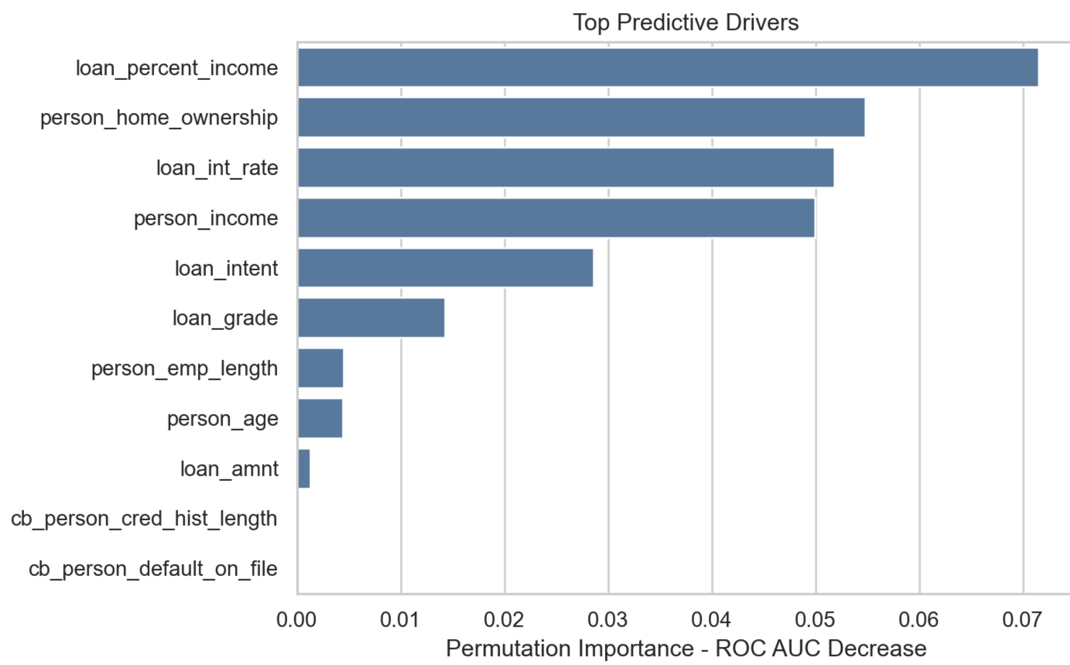
Test ROC-AUC	Test PR-AUC	Tuned test F1
0.954	0.872	0.808

At the selected threshold, the model identifies 1,215 defaults and misses 455. It produces 123 false positives, which should be interpreted as the cost of additional default capture.

7. Model Interpretation

Conclusion first: permutation importance confirms that the model's strongest drivers align with the EDA story. The model is primarily learning affordability, pricing, housing stability, income, and loan purpose.

Rank	Feature	Importance	Interpretation
1	loan_percent_income	0.0785	Affordability and repayment burden
2	loan_int_rate	0.0558	Pricing and risk signal
3	person_home_ownership	0.0514	Housing stability
4	person_income	0.0499	Repayment capacity
5	loan_intent	0.0288	Purpose-specific risk profile
6	loan_grade	0.0126	Underwriting risk tier



8. Recommendations and Submission Notes

The modeling results are directionally consistent with the EDA findings and demonstrate stable performance across cross-validation and holdout sets. The primary business consideration is how aggressively to set the operating threshold, depending on risk tolerance and review capacity.

- Select Gradient Boosting as the final model, as it achieves the strongest CV PR-AUC and robust out-of-sample performance.
- Use the model primarily as a risk-ranking tool, rather than a strict yes/no decision rule.
- Create risk bands based on predicted probabilities to align underwriting intensity with estimated borrower risk.
- Prioritize enhanced review for borrowers with:
    - High loan-to-income ratios
    - Higher interest rates
    - Riskier loan grades
    - Prior default history

Before production deployment, confirm that loan_grade and loan_int_rate are available at scoring time. If not, train a pre-underwriting variant excluding post-pricing features.

AI Transparency Statement

I used ChatGPT (and Codex-style AI assistance) to accelerate development, including generating initial code scaffolding, organizing parts of the modeling pipeline, and assisting with formatting. The overall analytical framework, EDA design, modeling strategy, validation methodology, threshold strategy and interpretation of results were developed and finalized by me, Juana Zhang.